

G graph, n vertices, m edges $\Rightarrow \omega(G) \geq n-m$ (*)

Q: Any connected graph with n vertices has at least $n-1$ edges.

$$\omega(G) = 1$$

$$(*) \quad 1 \geq n-m \Rightarrow m \geq n-1 \quad \checkmark$$

Theorem: G a simple graph with n vertices.

If the degree of each vertex is at least $\frac{n-1}{2}$ then G is connected.

Proof by contradiction:

assume $\sim$ of conclusion $\approx G$ is not connected

G has at least two components, $C_1, C_2, \dots$

let C_1 be the smallest component
(has min # of vertices)

$\Rightarrow C_1$ has at most $\frac{n}{2}$ vertices

Take any vertex u in C_1

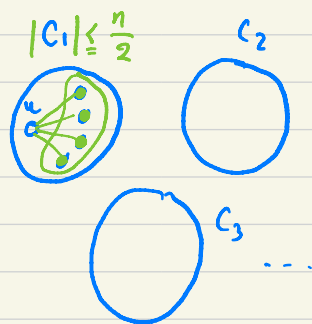
u can only be connected to vertices in C_1

$$d(u) \leq |C_1| - 1 \leq \frac{n}{2} - 1 = \frac{n-2}{2} < \frac{n-1}{2}$$

$\downarrow$

this contradicts the fact that each vertex (including u)

has degree $\geq \frac{n-1}{2}$. $\square$

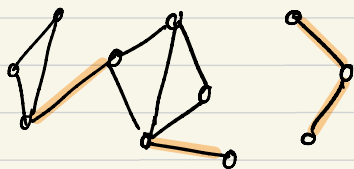
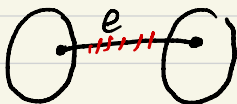


H.W. G , simple graph, n vertices.

Prove that if G has at least $\frac{(n-1)(n-2)}{2} + 1$ edges then G is connected.

Bridge (cut edge): is an edge e s.t. its removal increases the $\#$ of components.

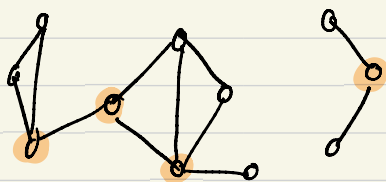
$$w(G-e) > w(G)$$



how many cut edges?

Cut vertex: is a vertex v s.t. its removal increases the $\#$ of components.

$$w(G-v) > w(G)$$



how many cut vertices?

cut vertex has degree ≥ 2

Theorem: If G is a connected graph with no odd vertices,
then G has no bridges.

Proof by contradiction.

assume $\sim$ of conclusion G has at least one bridge, (u, v)

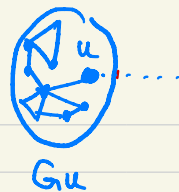
Let G_u, G_v be the components obtained after removing (u, v)



Let consider G_u

The degree of u in G_u is odd

all other vertices of G_u have even degrees.

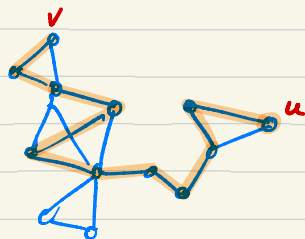


$\Rightarrow G_u$ has exactly one vertex of odd degree.

This contradicts the fact that the # of odd vertices is even. $\square$

Theorem: Any connected graph, with at least 2 vertices, has at least 2 vertices that are not cut vertices.

Proof: Consider a longest path in the graph
(max # of edges)

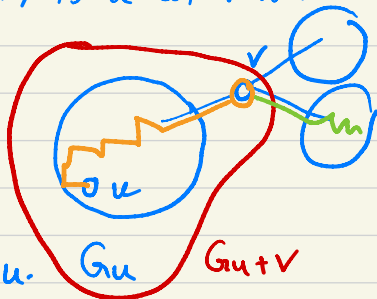


Claim: the two endpoints of this path are not cut vertices.

Prove claim by contradiction.

assume one of the endpoints, say v , is a cut vertex.

If we remove v , then we get at least two components.

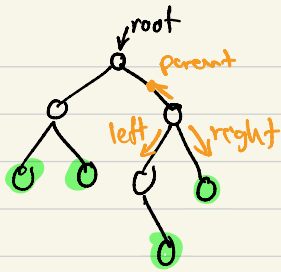


Let G_u be the component containing u .

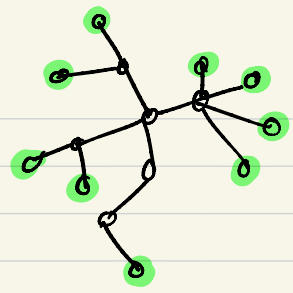
the longest path is in G_{u+v}

From v we can go to another component to make the path longer
this contradicts our assumption of longest path $\square$

tree: a connected graph with no cycle.



binary tree (Data Structure)



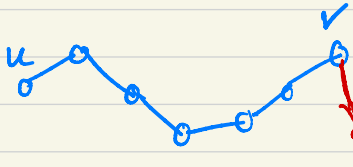
leaf: a degree-1 vertex

Theorem: Any tree (with ≥ 2 vertices) has at least 2 leaves.
degree-1 vertex

Proof: take a longest path $u \sim v$ in the tree

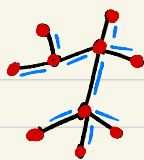
both u and v must have degree-1, because if one of them,

say v , has degree larger than 1, then we can make path longer. $\square$



Theorem: Any tree with $|V|$ vertices has $|E| = |V| - 1$ edges.

Proof by induction (on $|V|$)



$$|V| = 10$$

$$|E| = 9$$

Base Case ($|V| = 1$)

$$|V| = 1$$

$$|E| = |V| - 1 = 1 - 1 = 0 \quad \checkmark$$

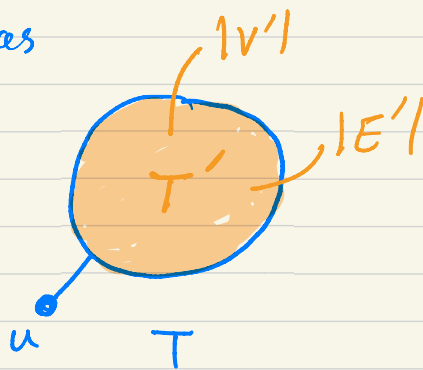
I.H.: assume any tree with $|V'|$ vertices ($|V'| < |V|$) has $|E'| = |V'| - 1$ edges.

I.S. Prove the claim for tree T with $|V|$ vertices

$$\downarrow$$
$$(|E| = |V| - 1)$$

Take a leaf u in T

Remove u to obtain T'



$$|V'| < |V|$$

By I.H. $|E'| = |V'| - 1$

$$|V'| = |V| - 1$$

$$|E'| = |E| - 1$$

$$|E| - 1 = |V| - 1 - 1$$

$$|E| = |V| - 1 \quad \checkmark$$

